Coordination of Volunteer Tutors in Vulnerable Educational Communities: Design, Compatibility Algorithm, and Simulation-Based Validation

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***Abstract*—In many schools serving low-income communities in Bogotá D.C., the risk of student dropout is worsened by the lack of personalized academic support: a supply of willing volunteer tutors exists, yet the coordination between volunteers, institutions, and students remains manual, informal, and difficult to scale, resulting in inefficient assignments and inconsistent coverage. This paper proposes a tutor-volunteer coordination system grounded in a weighted compatibility algorithm whose parameters were calibrated from a survey administered to 50 prospective volunteers—students and instructors at Universidad Distrital Francisco José de Caldas and Colegio Manuel Cepeda Vargas IED Sede B JM. Validation through discrete-event simulation and cellular automata showed that the algorithm achieves mean compatibility scores between 0.860 and 0.887, attains 100 % coverage of active projects with a minimum pool of 30 volunteers, and produces an 18 % relative increase in community engagement as an emergent behavior of the volunteer social network.**

***Index Terms*—educational volunteering, compatibility algorithm, discrete-event simulation, cellular automata, systems analysis, vulnerable communities, Bogotá**

I. Introduction

In schools serving low-income communities in Bogotá D.C., student dropout is a structural problem driven by multiple reinforcing factors, among them the absence of timely and personalized academic support [7]. This gap became apparent to our team through conversations with fellow university students who reported a lack of structured academic assistance, an observation later confirmed by the survey data. Although a sizeable population of university students is willing to provide tutoring on a voluntary basis—as shown by the survey, in which nearly 78 % of respondents expressed willingness to do so—the connection between this supply and the actual demand of educational institutions is poorly organized or insufficiently far-reaching: schedules are negotiated manually and no mechanism exists to verify that a tutor is qualified for the specific subject area a student needs. The result is a system that wastes available talent while leaving concrete educational needs unmet.

Volunteer coordination in social contexts has been studied from a variety of perspectives. In the field of humanitarian logistics, resource-allocation models based on linear programming and heuristics have demonstrated substantial improvements in coverage and response times [4]. In the domain of social networks and complex systems, agent-based models and cellular automata have proven capable of capturing social contagion dynamics—both positive, such as the spread of commitment, and negative, such as burnout—that are not detectable through component-level analysis [5], [6]. From a systems-engineering standpoint, we draw on the ISO 9001:2015 quality framework [2] and the ISO 31000:2018 risk-management guidelines [1] to design a robust sociotechnical system, while PMBOK 7 [3] structures its planning and execution.

This paper describes the complete systems-engineering cycle applied to this problem—from primary data collection and analysis of the current system through algorithm design, risk management, and computational validation using two complementary simulation paradigms. The central contribution is to demonstrate, with quantitative evidence, that a well-designed coordination system can transform an inefficient, informal process into one that is traceable, scalable, and capable of producing measurable impact on educational coverage in vulnerable communities.

II. Methods and Materials

A. System Analysis and Data Collection

The analysis began with a characterization of the existing volunteer system—understood as the set of actors, processes, and interactions that currently sustain, in a fragmented way, academic support in Bogotá's vulnerable communities. The actors identified were: university volunteers (the supply side of tutoring services), educational institutions at the secondary and post-secondary levels (the demand side), and students (the final beneficiaries). Key environmental constraints that limit system performance include limited digital connectivity in certain areas, rigid institutional academic calendars, fragmented volunteer availability, and socioeconomic barriers affecting both tutors and students.

To quantify the characteristics of the prospective volunteer population, a digital survey was designed and administered to 50 students at Universidad Distrital Francisco José de Caldas, distributed through academic groups and student volunteer networks. The survey captured information on field of study, academic competencies for teaching, prior tutoring experience, weekly time availability, and preferred delivery mode (in-person, virtual, or hybrid). The findings most relevant to system design were: 40 % of respondents belong to Engineering programs, 22 % to Education programs; 54 % report fewer than two hours per week available for volunteering; and in-person and virtual modalities tied in preference at 36 % each, while 28 % expressed willingness to work in both.

B. System Architecture

1) Stakeholders and External Actors: The system architecture integrates stakeholders and operational processes through a three-tier structure that ensures coherence in volunteer management. Volunteers interact through a dedicated portal, with students being the primary beneficiaries, who can track their learning progress and session history in detail. Institutions manage projects and collaboration schedules through the institutional portal, while the external environment is supported by the Google Sheets API, which serves as the temporary persistence layer for the platform's data.

Operational processes are executed through three well-defined logical tiers. The registration and management tier handles the onboarding of volunteers and institutions. The intelligent coordination tier runs the compatibility algorithm and the conflict-free scheduling engine. In parallel, a tracking and analytics tier continuously monitors academic impact and generates reports, supported by an automated notification service that ensures effective communication across actors.

Regarding information flows, the system receives inputs such as volunteer profiles, schedule availability, project requirements, and session records. These data are processed through a continuous exchange between the presentation and application tiers, producing outputs consisting of optimized assignments, alerts, notifications, and detailed impact reports. Interactions between organizational and technological components are central to system operation: the user interface in the presentation tier acts as a bridge through which stakeholders translate their organizational needs into data processable by the application tier, which in turn ensures full alignment between institutional requirements and actual volunteer availability. The data tier functions as institutional memory, ensuring that compatibility and scheduling decisions remain aligned with academic objectives.

In terms of inputs, outputs, and constraints, the system processes profiles, project data, and activity records to produce consolidated schedules, impact metrics, and volunteering account summaries. The system operates under the technical constraint of using Google Sheets as its data repository and maintains a quality commitment grounded in strict adherence to the modularity and fault-tolerance principles defined by ISO 9001:2015.

2) Risk Management: The risk analysis followed the ISO 31000:2018 methodology [1] and produced two registers: technical and operational. Among the critical risks (high probability $\times$ high impact), the analysis identified poor data quality entered by volunteers and institutions (RT01) and fluctuation in volunteer availability (RO01). For RO01—the most relevant to the simulation results—a critical threshold was established at a minimum pool of 30 active volunteers, below which project coverage degrades rapidly. Mitigation measures include an emergency dissemination protocol through university academic networks and a waiting-list system with temporary assignments.

It is also worth noting that the reliance on Google Sheets as a temporary database introduces risks of data saturation and integrity limitations as data volume grows. Contingency plans include documented migration to PostgreSQL and manual schedule validation procedures in the event of algorithmic errors.

C. Compatibility Algorithm

The core of the system is the compatibility function $C(v, p)$, which computes a composite score for each volunteer–project pair:

$$\begin{aligned} C(v,p) = 0.40 \cdot Sk(v,p) + 0.30 \cdot Sc(v,p) \\ + 0.15 \cdot Lc(v,p) + 0.15 \cdot Ex(v,p) \end{aligned} \tag{1}$$

where Sk measures the overlap between the volunteer's academic competencies and the curricular needs of the project; Sc quantifies the percentage of required time slots that the volunteer can cover; Lc encodes geographic proximity for in-person modalities or connectivity quality for virtual ones; and Ex maps the volunteer's prior teaching experience to the grade level targeted by the project.

The weights were derived from the survey findings. The highest weight ($w_{Sk} = 0.40$) reflects that 40 % of volunteers come from Engineering programs and that the partner institutions prioritize support in mathematics and technology—a natural alignment between supply and demand that the system should capitalize on. The schedule weight ($w_{Sc} = 0.30$) is justified by the finding that 54 % of volunteers have limited availability (fewer than two hours per week), making schedule coordination the second most critical bottleneck in the current system. The weights for location and experience ($w = 0.15$ each) reflect their relevance as secondary factors compared to academic and temporal compatibility. A minimum acceptance threshold of $C(v, p) > 0.80$ was established for any assignment.

D. Simulation Models

The campus simulation model is structured over a 30×30 grid representing 900 community members, and operates as a feedback system in which collective behavior is the central analytical dimension. A fixed random seed (value 42) is used throughout to ensure full reproducibility under identical input conditions.

The simulation is initialized with 8 % of the population in the active-volunteer state, which serves as the critical mass required for positive social contagion to take hold. From that starting point, the system manages a five-state life cycle through which individuals transition between inactivity, awareness of the platform, engagement, active tutoring, and burnout. The dynamics are governed by two competing forces.

The first is the inflow force, driven by local social influence within the Moore neighborhood and amplified by the digital platform. When the platform is active, it significantly increases recruitment efficiency by facilitating transitions from the aware state to the engaged state. The second is the outflow force, defined by burnout contagion: the probability of a volunteer becoming burned out grows as a function of the number of nearby neighbors already in that state, creating saturation clusters that stall activity in specific areas of the grid.

From a systems perspective, the simulation demonstrates that operational stability does not depend solely on massive recruitment, but on the dynamic equilibrium between the inflow rate and the recovery rate of burned-out volunteers re-entering the cycle as inactive. If burnout propagation outpaces the platform's ability to attract new volunteers, the system loses its critical mass and collapses. The model thus provides a tool for identifying tipping points where depletion pressure exceeds the management's pull capacity, enabling preventive adjustments to volunteer density or platform activation intensity before the network destabilizes.

Table I
SIMULATION PARAMETERS CALIBRATED FROM THE SURVEY AND RISK ANALYSIS

Parameter	Value	Source
Low-availability ratio	54 %	Survey Q5
Engineering volunteers	40 %	Survey Q2
Session cancellation rate	15 %	Risk RO03
Mean algorithm latency	45 ms	NFR target
Skill weight w_{Sk}	0.40	Survey
Schedule weight w_{Sc}	0.30	Survey
Simulation horizon	12 weeks	Project plan
CA grid size	30×30	UD campus
Initial engagement ratio	8 %	Survey baseline

III. RESULTS AND DISCUSSION

A. Discrete-Event Simulation

Table II summarizes the quantitative results across the three simulated scenarios. All three scenarios achieved 100 % project coverage and mean compatibility scores above the 0.80 threshold, validating the design assumptions established in Workshop 2. Algorithm latency remained at approximately 45 ms across all tested configurations, well below the 60-second non-functional requirement.

An unexpected finding emerged from the Stress scenario: the mean compatibility score was the highest of the three (0.887), surpassing even the Optimistic case. This *quality paradox under scarcity* has a direct explanation in the algorithm's optimization logic: with fewer volunteers available, each assignment concentrates on the most compatible pair rather than distributing load to marginally qualified volunteers. This result was not anticipated in the original design and constitutes an emergent property of the system that can inform operational policy: under shortage conditions, the system should prioritize match quality over coverage breadth.

Table II
DISCRETE-EVENT SIMULATION RESULTS

Scenario	Coverage	$\bar{C}(v,p)$	Cancelled	Target met?
Baseline (50 vol.)	100 %	0.860	3	**Yes**
Optimistic (80 vol.)	100 %	0.870	4	**Yes**
Stress (25 vol.)	100 %	0.887	1	**Yes**
Design target		Coverage >70 %, $C(v,p) > 0.80$		

Sensitivity analysis confirmed three additional design decisions. First, algorithm latency scales sub-linearly with pool size, remaining below 60 s for all tested sizes (10 to 100 volunteers). Second, coverage degrades rapidly when the volunteer count falls below 30, empirically validating the RO01 risk threshold. Third, the weight $w_{Sk} = 0.40$ lies within an optimal plateau spanning 0.35 to 0.50 on the sensitivity curve; values above 0.55 degrade mean quality by underweighting the schedule and location factors.

B. Cellular Automata: Emergent Behaviors

Each cell in the 30×30 grid represents a member of the university community. The grid symbolically represents the UDFJC campus as a population of 900 individuals who may or may not become volunteers. There is no global communication between cells: each person "sees" only their eight immediate neighbors. From this purely local rule set, the global behavior of the system emerges.

This modeling choice was deliberate: volunteer recruitment does not operate through mass broadcasting but through close social influence—a person is more likely to participate when they observe that classmates or close peers are already doing so. That mechanism is precisely what the cellular automaton captures.

The model revealed two principal emergent phenomena. The first is a *network amplification effect*: with the platform active, total engagement reached 24.4 % by week 20, compared to 20.7 % without it—an 18 % relative increase. This gain is not encoded in any individual cell rule; it emerges from the compounding of small probability increments propagating across the social grid. Critical mass formation was observable by week 8, when platform-enabled conditions showed dense clusters of active tutors.

The second is *burnout contagion*, which was not identified in any of the risk registers produced prior to the simulation phase. When burned-out volunteers cluster spatially—a pattern that arises when high-demand projects concentrate in the same geographic zone—the burnout risk for neighboring active volunteers increases in cascade, consistent with Turing morphogenesis dynamics in complex systems [5]. This finding led to the proposal of a new operational risk (RO05) and a session-load monitoring module that triggers alerts when a tutor exceeds three sessions per week.

IV. Conclusions

This paper presented the complete systems-engineering cycle applied to the coordination of volunteer tutors in vulnerable educational communities in Bogotá D.C. Analysis of the current system identified three structural bottlenecks that account for the inefficiency of the manual process: information asymmetry between tutors and institutions, the absence of competency validation, and schedule fragmentation. From that diagnosis, the compatibility algorithm $C(v, p)$ was designed with weights grounded directly in empirical evidence and validated through two complementary simulation paradigms that confirmed fulfillment of all pre-established design targets.

The most significant contribution of the simulation process was the identification of burnout contagion as an emergent risk invisible to component-level analysis—a finding we did not anticipate when the project began, and one that reshaped how we think about the operational risks of sociotechnical systems. It demonstrates the value of behavior-oriented simulation, specifically cellular automata, as an analytical tool for sociotechnical systems beyond its traditional applications in physical or computational domains.

As future work, we propose conducting a real pilot with two or three partner educational institutions in Bogotá to recalibrate the algorithm weights using post-session satisfaction data, integrating the burnout monitoring module into the scheduling engine, and analyzing the geographic distribution of projects to prevent the burnout clusters identified in simulation.

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